Eye Trauma

TB-11 | Trauma & Burns

Quick flow

01 Scene + Life Threats [EMT / AEMT / Paramedic]

PPE - ABCs - associated head/facial trauma - spinal motion restriction only when indicated

02 Chemical Exposure? [EMT / AEMT / Paramedic]

Ordinary splash: start immediate gentle copious irrigation through handoff. Known/strongly suspected open globe with contamination: no irrigation into globe or pressure; noncontact rigid shield and urgent specialist/Medical Control guidance.

03 Thermal Eye Injury? [EMT / AEMT / Paramedic]

Briefly cool or gently irrigate with room-temperature water or saline - protect the eye - do not use ice

04 Penetrating / Open Globe? [EMT / AEMT / Paramedic]

No pressure, manipulation, irrigation, drops, or object removal - rigid shield - stabilize protruding object

05 Blunt / Superficial? [EMT / AEMT / Paramedic]

Assess vision and red flags - shield if globe injury suspected - irrigate only loose debris

06 Vision Threat? [EMT / AEMT / Paramedic]

Vision loss - irregular pupil - hyphema - proptosis - extrusion - severe pain - high-velocity mechanism

07 Protect + Transport [EMT / AEMT / Paramedic]

Control pain/nausea within scope - minimize Valsalva - serial vision check - early specialty notification

Overview

- Use for blunt, penetrating, thermal, or chemical eye injury. Treat life threats and associated trauma first, then protect vision by preventing pressure on a potentially open globe and beginning immediate irrigation for an ordinary chemical splash; known or strongly suspected open globe with contamination requires the specific exception below.
- Any vision loss, irregular pupil, hyphema, extrusion of ocular contents, impaled object, marked proptosis, severe pain, or high-velocity mechanism requires urgent transport and early receiving-facility notification.

Provider-Level Actions

Protect the injured eye, preserve baseline findings, treat pain and nausea, and avoid transport delay.

EMT

- Treat immediate airway, breathing, circulation, hemorrhage, and associated trauma threats. Use spinal motion restriction only when indicated by mechanism and examination.
- For an ordinary chemical splash, begin immediate gentle copious water or normal-saline irrigation and continue during transport through handoff; do not wait for product identification or ideal fluid. Remove contact lenses only when easily accomplished without delaying irrigation. Gently cool or irrigate thermal exposure as directed below.
- For suspected penetrating or open-globe injury, do not apply pressure, remove an object, manipulate the eye, irrigate, or instill medication. Stabilize a protruding object without pressure and place a rigid eye shield when available.
- If an open globe is known or strongly suspected with chemical contamination, do not irrigate into the globe or apply pressure. Protect with a noncontact rigid shield and urgently obtain ophthalmology, Poison Help, and Medical Control guidance. A hypothetical concern for rupture must not delay ordinary splash irrigation.

- Assess and document visual acuity in each eye separately when feasible, pupils, visible injury, mechanism, pain, and changes during care. Do not delay time-sensitive transport to complete testing.

AEMT

- Perform all EMT actions plus vascular access when clinically indicated and authorized analgesia or antiemetic treatment under the current standing orders.
- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.
- Monitor for deterioration from associated head/facial trauma, hemorrhage, chemical exposure, or systemic injury and support transport without delaying irrigation or eye protection.
- Avoid medications or interventions that increase agitation, vomiting, coughing, or pressure on a suspected open globe.

Paramedic

- Perform all prior actions; lead advanced airway, analgesia, antiemetic, monitoring, Medical Control, and destination decisions when the injury or associated trauma is severe.
- Use early Medical Control and receiving-facility consultation for vision loss, suspected open globe, impaled object, orbital compartment warning signs, hydrofluoric-acid exposure, or uncertain destination.
- Do not use ketamine for analgesia in suspected penetrating or open-globe eye trauma under this protocol unless Medical Control expressly authorizes it.

Indications

- Pain, photophobia, redness, swelling, inability to open the eye, abnormal pupil, hyphema, proptosis, diplopia, or loss of vision after injury.

Assessment | Focused eye assessment

- Identify the exact mechanism, time, eye involved, chemical or product when applicable, contact-lens use, protective eyewear, high-velocity metal/glass exposure, and associated head or facial trauma.
- Assess visual acuity in each eye separately before treatment when feasible and after treatment when it will not delay care. If a chart is unavailable, document the ability to read, count fingers, detect hand motion, or perceive light.
- Inspect without pressure for pupil shape and reactivity, redness, bleeding, hyphema, extrusion of tissue or fluid, corneal clouding, foreign body, lid injury, proptosis, and peri-orbital swelling.
- Assess extraocular movement and diplopia only when it can be done gently and without manipulating a suspected open globe. Do not force the eyelids open.
- Reassess pain, vision, pupils, mental status, and associated trauma throughout care.

Assessment | Vision-threatening findings

- Decreased or lost vision, irregular or teardrop pupil, visible penetration, extruded ocular contents, hyphema, or an impaled object.
- Marked proptosis, rapidly increasing orbital swelling, severe pain, decreased vision, or an afferent pupillary abnormality after blunt trauma.
- Chemical burns, especially alkali or hydrofluoric-acid exposure; high-velocity metal or glass; explosion; severe lid laceration; or inability to adequately examine the eye.

Treatment steps

- Ordinary chemical splash: begin immediate gentle copious water or normal-saline irrigation and continue during transport through handoff; do not wait for product identification or ideal fluid. Remove contact lenses when easily possible, irrigate from the inner corner outward, and prevent runoff into the unaffected eye or onto responders.
- When trained and equipped to use an agency-approved ocular-pH endpoint, follow that endpoint and reassess pH after a pause. When those resources are unavailable, continue ordinary chemical-splash irrigation through handoff; do not delay irrigation to obtain equipment.
- Brush away a dry chemical before irrigation when safe and appropriate. Do not attempt acid-base neutralization. Follow TE-09 Chemical / HazMat Exposure and Poison Help guidance for unusual or water-reactive agents.
- Suspected penetrating/open-globe injury: do not remove an impaled object, manipulate the eye, apply pressure, irrigate, or instill medication. Stabilize a protruding object with bulky material around it without pressure, then protect with a rigid shield.
- Loose superficial debris may be irrigated only when penetrating injury or globe rupture is not suspected. Do not remove an embedded foreign body or use cotton swabs, magnets, or instruments on the eye.
- Provide authorized analgesia and antiemetic care, minimize coughing, vomiting, agitation, and Valsalva, and position with the head elevated when safe for the overall trauma condition.
- Transport promptly to the most appropriate facility under the trauma and specialty-destination plan. Notify early for open globe, impaled object, vision loss, orbital-compartment warning signs, significant chemical burn, or severe associated trauma.
- Document baseline and repeat vision, pupil findings, laterality, mechanism, product/exposure time, irrigation start and duration, eye protection, medication, response, consultation, and destination rationale.

Medications

Analgesic

Use the current Claiborne County pain-management protocol and provider-level standing orders.

- Titrate carefully and monitor airway, ventilation, mental status, and blood pressure.

Antiemetic

Use the current age-appropriate Claiborne County standing order when nausea or vomiting threatens the injured eye.

- Preventing vomiting and Valsalva is especially important when open-globe injury is suspected.

Topical ophthalmic anesthetic

Only if specifically stocked and adopted in a finalized Claiborne County standing order or ordered by Medical Control.

- May facilitate irrigation when authorized; never dispense for patient home use.

Warnings / precautions

- Never apply pressure to an eye with trauma, suspected penetration, irregular pupil, hyphema, extrusion, or possible globe rupture.
- Do not remove an impaled or embedded object and do not attempt field repair of an eyelid or globe injury.
- Do not delay ordinary chemical-splash irrigation to identify the product, obtain visual acuity, locate saline, or contact Poison Help. The known/strongly suspected contaminated open-globe exception requires protection and urgent expert guidance, not irrigation into the globe.
- Do not neutralize a chemical on the eye and do not use ice, ointment, cotton swabs, magnets, or instruments.
- Do not patch either eye. For suspected open-globe injury, protect the affected eye with a rigid shield without pressure.

- Sudden painless non-traumatic vision loss is not eye trauma. Contact Medical Control and use the appropriate medical pathway; do not apply ocular pressure when trauma or globe rupture is possible.

Clinical pearls

- A normal external appearance does not exclude an open globe or intraocular foreign body after a high-velocity mechanism.
- Pediatric patients may describe vision poorly and may resist examination; use calm, age-appropriate assessment and avoid forced eyelid opening.

Special populations | Pediatric patient

- Use a parent or caregiver to help with reassurance and irrigation when safe.
- Measure and record weight with the length-based system when age/size appropriate before medication, but do not delay irrigation or transport.
- Suspect non-accidental trauma when the history, development, mechanism, or injury pattern is inconsistent; follow the Family Violence / Abuse protocol.

Related pathways / contacts

Chemical / HazMat Exposure - Responder safety, decontamination, agent cards, and Poison Help.

Chemical and Electrical Burn - Associated facial, skin, electrical, or thermal burn care.

Head Trauma - Associated head injury and serial neurologic assessment.

Multiple Trauma - Multisystem injury and trauma destination decisions.

Source and references

Tennessee EMS ALS/BLS Blended Protocol Guidelines 2024-2025 - SOP #405 Eye Trauma. | Tennessee EMS ALS/BLS Blended Protocol Guidelines 2024-2025 - Ketamine Reference. | Claiborne County EMS Chemical / HazMat Exposure, Head Trauma, Multiple Trauma, and pain-management protocols.

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